Math 372: Practice Final Exam #1

Instructions: You have 2 hours. No calculuators, no notes. You must show your work to recieve credit.

Problem 1. An email provider observes that 80% of emails are spam. It intends to filter spam emails before they reach the inbox. The filter's accuracy for detecting spam email is 98% and chances of tagging a non-spam email as spam is 4%.

(a) Draw a tree diagram to illustrate this situation

(b) If a certain email is tagged as spam, find the probability that it is not a spam email. That is, find

$$\mathbb{P}\left[\text{email is not spam} \mid \text{email is tagged as spam}\right]$$

Problem 2. Let A and B be events for which you know $\mathbb{P}[A \cup B] = 0.57$, $\mathbb{P}[A] = 0.22$, and $\mathbb{P}[A \cap B] = 0.16$.

(a) Find $\mathbb{P}[B]$

(b) Are A and B independent? Justify your answer.

(c) Are A and B disjoint? Justify your answer.

(d) Find $\mathbb{P}[A \cap B^c]$

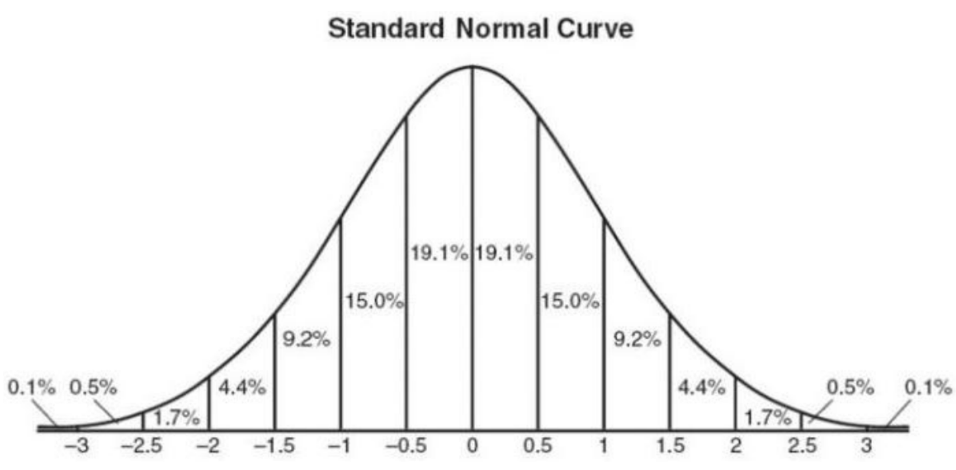
(e) Find $\mathbb{P}[A \mid A \cup B]$

(f) Find the probability that exactly one of the two events A, B occurs.

Problem 3. The nation of Oceania has always been at war with Eurasia. To establish peace, Oceania's leadership has unanimously authorized a preemptive strike against 48 pre-selected military cities in Eurasian territory. Military analysts at the Ministry of Peace estimate that each target has a probability of 3/4 of being destroyed in this first strike.

The operation will be regarded as a strategic success under Oceania's doctrine of peace through calibrated destabilization if at least 30 targets are destroyed. Using the Central Limit Theorem, estimate the probability that the first strike achieves this benchmark.[1]

Hints: The following may be useful $\sqrt{48} = 4\sqrt{3}$. Also, the graph of the standard normal pdf is



[1]Responses deviating significantly from 100% confidence will be flagged for review by the Ministry of Truth.

Problem 4.

Morkbork, an alien from the star system Dihedral-5, is testing out the new “magic 9-ball” that she recently purchased at an interstellar gas station. Here’s how it works: if she shakes the 9-ball, it returns one of the following 9 alien symbols:

$$\Phi \quad \sim \quad \nu \quad \forall \quad \otimes \quad \eth \quad \because \quad \xi \quad ϶$$

Morkbork would like to know whether or not the 9 symbols are equally likely, so naturally she decides to perform a chi-squared goodness-of-fit test. To generate some data, she shakes the magic 9-ball 90 times and records the number of times each symbol is returned. She summarizes her data in the following table:

symbol	Φ	$\sim$	ν	$\forall$	$\otimes$	$\eth$	$\because$	ξ	϶
observed count	9	14	6	13	12	8	11	8	9

(a) Help Morkbork get started by stating an appropriate null and alternative hypothesis.

(b) Assuming Morkbork’s null hypothesis is true, what are the expected counts? Write your answer as a table.

(c) Compute the chi-squared statistic χ^2 for Morkbork’s data.

(d) Circle the correct phrase to complete the sentence: *A larger chi-squared value means the observed data is (very different from / very similar to) the expected data.*

(e) Fill in the blank: *In this problem, the chi-squared statistic χ^2 has a chi-squared distribution with* ______ *degrees of freedom.*

(f) In plain English, describe in one or two compete sentences what p-value means.

(g) After consulting a statistics textbook, Morkbork determines that the chi-squared statistic has pdf

$$f(x) = \begin{cases} \frac{1}{96}x^3e^{-x/2} & : \quad x \geq 0 \\ 0 & : \quad x < 0 \end{cases}$$

Use this formula to set up an integral to compute the p-value for your answer to part (c). **Do not evaluate the integral.**

(h) Morkbork computes the p-value to be $p = .69$. What conclusion can she draw from this result? Please explain your answer in one or two sentences.

Problem 5. Two cards are randomly drawn from a deck of cards. Let A be the event that at least one ace is drawn. Let A_s be the event that the ace of spades is chosen. And let B be the event that both cards are aces. Compute the following conditional probabilities. You do not need to simplify your answers.

(a) $\mathbb{P}[B \mid A_\text{s}]$

(b) $\mathbb{P}[B \mid A]$

(c) $\mathbb{P}[A_\text{s} \mid B]$

Problem 6 (continuous rv)**.** A continuous random variable X has the following pdf:

$$f(x) = \begin{cases} 4x^{-5} & : \quad x \geq 1 \\ 0 & : \quad x < 1 \end{cases}$$

(a) Verify that f is a valid pdf

(b) Find the cumulative density function $F(x)$

(c) Consider a game with a \$2 entry fee. The game pays out X dollars. Find the mean and variance of your net winnings.

Now consider playing the game repeatedly, using independent observations of X. Find the following

(d) Find the probability that the payout X is greater than \$10.

(e) What is the probability that the largest payoff in the first six plays does not exceed \$10?

(f) What is the probability that exactly three of the first six payoffs exceed \$10?

Problem 7. Consider the following four datapoints:

$$(-1, 2), \quad (0, 0), \quad (1, -3), \quad \text{and} \quad (2, -5)$$

(a) Find the best straight-line fit to the data.

(b) Plot points and the line.